

# Zazu Personal Manager

A React Native + Expo application with SQLite local database storage, biometric authentication, backup/export functionality, and support for future database migrations.

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## Prerequisites

Install:

- Node.js LTS
- Git
- Expo CLI
- EAS CLI

Verify installation:

```
node -v  
npm -v
```

Install EAS CLI:

```
npm install -g eas-cli
```

Login:

```
eas login
```

---

## Project Setup

Clone repository:

```
git clone <repository-url>  
cd ZazuPersonalManager
```

Install dependencies:

```
npm install
```

Install Expo dependencies:

```
npx expo install
```

Verify project health:

```
npx expo-doctor
```

---

## Running The Application

Start Expo:

```
npx expo start
```

Android:

```
npx expo run:android
```

iOS:

```
npx expo run:ios
```

---

## SQLite Database Architecture

Database file:

```
app.db
```

Open database:

```
import * as SQLite from 'expo-sqlite';  
  
export const db = await SQLite.openDatabaseAsync('app.db');
```

Database location:

Android:

```
/data/data/<package-name>/databases/
```

iOS:

```
Application Sandbox/Documents/
```

Important:

Updating the application does NOT remove the database.

Uninstalling the application WILL remove the database.

---

## Database Versioning

Create metadata table:

```
CREATE TABLE IF NOT EXISTS meta (  
  key TEXT PRIMARY KEY,
```

```
    value TEXT
  );
```

Store version:

```
INSERT OR REPLACE INTO meta(key,value)
VALUES('db_version','1');
```

---

# Database Migration System

## Migration Structure

```
src/
├── database/
│   ├── migrations/
│   │   ├── 001_initial.ts
│   │   ├── 002_health_table.ts
│   │   └── 003_asset_table.ts
│   ├── migrationRunner.ts
│   └── database.ts
```

---

## Migration Interface

```
export interface Migration {
  version: number;
  migrate: (db: SQLite.SQLiteDatabase) => Promise<void>;
}
```

---

## Example Migration

### 001\_initial.ts

```
export default {
  version: 1,
  migrate: async (db) => {
    await db.execAsync(`
```

```
CREATE TABLE IF NOT EXISTS users (  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  name TEXT  
);  
`);  
}  
};
```

---

## 002\_health\_table.ts

```
export default {  
  version: 2,  
  migrate: async (db) => {  
    await db.execAsync(`  
      CREATE TABLE IF NOT EXISTS health_entries (  
        id INTEGER PRIMARY KEY AUTOINCREMENT,  
        date TEXT,  
        steps INTEGER,  
        calories INTEGER  
      );  
    `);  
  }  
};
```

---

## Migration Runner

```
export async function runMigrations(db) {  
  const currentVersion = await getCurrentVersion(db);  
  
  for (const migration of migrations) {  
    if (migration.version > currentVersion) {  
      await migration.migrate(db);  
      await setVersion(db, migration.version);  
    }  
  }  
}
```

## Application Startup

```
await runMigrations(db);
```

This guarantees every future release automatically upgrades the database safely.

---

## Preserving User Data During Updates

### Safe

Application Update:

```
v1 APK → v2 APK
```

Database remains intact.

User data remains intact.

---

### Unsafe

Application Uninstall:

```
User removes app
```

Database removed.

All local data lost.

---

## Requirements For Safe Updates

Keep identical:

```
{  
  "android": {  
    "package": "com.codingloo.zazu"
```

```
}  
}
```

Do NOT change package name.

Do NOT change bundle identifier.

---

## Android Build Process

### Configure EAS

Initialize:

```
eas build:configure
```

Creates:

```
eas.json
```

---

### Login

```
eas login
```

---

### Verify Project

```
npx expo-doctor
```

Fix all errors before building.

---

### Preview APK

Build installable APK:

```
eas build --platform android --profile preview
```

Download APK from:

<https://expo.dev/accounts/<account>/projects/<project>/builds>

---

## Production AAB

Google Play release:

```
eas build --platform android --profile production
```

Produces:

```
.aab
```

---

## iOS Build Process

Requirements:

- Apple Developer Account
- EAS Login

Build:

```
eas build --platform ios
```

or

```
eas build --platform ios --profile production
```

Produces:

```
.ipa
```

---



## Example eas.json

```
{
  "build": {
    "preview": {
      "android": {
        "buildType": "apk"
      },
      "env": {
        "NODE_ENV": "production"
      }
    },
    "production": {
      "env": {
        "NODE_ENV": "production"
      }
    }
  }
}
```

---

## Release Process

Version update:

```
version: "1.0.1"
```

Increment:

```
{
  "android": {
    "versionCode": 2
  }
}
```

```
{
  "ios": {
    "buildNumber": "2"
  }
}
```

Build:

```
eas build --platform all --profile production
```

---

## Backup & Restore Strategy

Recommended:

Export database:

```
app.db
```

Store backup:

- Device Storage
- Google Drive
- iCloud
- Dropbox

Restore:

Copy backup database into application database folder.

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## Troubleshooting

### Kotlin Version Error

Error:

```
Compose Compiler requires Kotlin version 1.9.25
```

Install:

```
npx expo install expo-build-properties
```

app.config.js:

```
[
  "expo-build-properties",
  {
    android: {
      kotlinVersion: "1.9.25"
    }
  }
]
```

---

## Missing Dependencies

Install:

```
npx expo install expo-constants expo-linking
```

---

## Clean Build

```
rm -rf node_modules
rm package-lock.json

npm install
```

Then:

```
npx expo prebuild --clean
```

---

## Verify Environment

```
npx expo-doctor
```

Expected:

```
All checks passed
```

---

# Future Database Migration Example

Version 1:

users

Version 2:

health\_entries

Version 3:

assets

When user upgrades:

v1 -> v3

Migration runner executes:

001  
002  
003

Automatically.

No user data loss.

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## Recommended Release Checklist

Before every release:

- Update version
- Increment versionCode
- Increment buildNumber
- Run migrations locally
- Test backup/restore
- Run expo-doctor

- Build preview APK
- Verify upgrade path
- Build production release

End of Document